Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Full-Stack Engineer with 3 years of in-house industry experience at HDFC Asset Management shipping web applications to real internal users end to end: ReactJS frontends served by Nginx, Python FastAPI backends over PostgreSQL, and LLM features (RAG on pgvector, tool-calling agents on Azure OpenAI) built into the workflows, with LDAP-based role access control and Docker releases through a Jenkins and GitLab CI/CD pipeline. Took each product from requirements through release and iteration with Compliance, Client Service and Investment stakeholders, rebuilt Power BI dashboards as React views for the wider organisation, and acted as de facto team lead for up to 4 interns; twice recognised internally with Best Team Project (2024, 2025). Recently completed an MSc in Data Science at the University of Bristol (result awaited). Seeking the Full-Stack Product Engineer role at Norton Rose Fulbright, building user-facing AI products for legal services.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Full-Stack Product Delivery: React, FastAPI & AI Features

- Designed and built internal **GenAI** web applications end to end for Compliance, Client Service and Investment: a **ReactJS** frontend served by **Nginx**, **Python FastAPI** microservices, and **RAG** (**LlamaIndex** with **Azure OpenAI GPT-3.5 & Llama-3**) over a separate **pgvector** database per department so access stayed siloed; initially the only engineer alongside my manager, later leading as the team grew.
- Built **agentic tool-calling** workflows where AI agents call **Python functions** as tools to run SQL-backed fund-comparison calculations and generate **PDF** reports on user request, with the same comparisons available directly in the UI without an AI call; built **ExpressJS** APIs for the calculations during development, guardrailed responses through **system prompts** and versioned every prompt change in **Git**.
- Released each application to real users, iterated on what they found, and had the **Compliance** department review every application before release; chose models by cost, with **Llama-3 self-hosted on a GPU EC2 instance** for the largest department (Client Service) and stronger **Azure OpenAI** outputs for decision makers, with the ability to switch models.
- Built a **Python** email-processing automation that turns over **300 daily emails** from credit rating agencies into structured data, fuzzy-matches them to tracked companies and sends analysts a daily summary, saving **60 manual analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images, which also improved accuracy.

Data-Heavy Interfaces, APIs & Integrations

- Developed **Power BI** dashboards for fortnightly credit report analysis, migrating financial analysts from **Excel** to Power BI and improving analyst efficiency by **10 hours a week**; recreated them in full in the **ReactJS** frontend so the wider organisation could use the data without Power BI licensing costs.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly (fixed income) and

monthly (equity), integrated with legacy data from analysts' Excel files; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.

Secure Deployment, Access Control & Team Delivery

- Designed the infrastructure for internal analytics platforms and GenAI applications on **AWS** utilizing core services such as **EC2, S3, ECR** and **Lambda**, with **Docker** containers for the **FastAPI** microservices, separate frontend and backend repositories, and a one-click **Jenkins** and **GitLab CI/CD** pipeline deploying to dev and prod.
- Implemented **RBAC** for the GenAI apps with a Python authentication function against the company **LDAP** server; restricted access to company private IPs using **AWS Security Groups**; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the **VAPT** (vulnerability assessment and penetration testing); coordinated with the security team on port whitelisting.
- Worked directly with stakeholders in **Client Service, Compliance and Investment** to turn requirements into working tools that replace manual work; acted as **de facto team lead for up to 4 interns** in my final year; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes (React with Vite and Tailwind, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). Designed the interface around slow, unreliable LLM calls: results stream to the UI over server-sent events as each finishes, partial results are salvaged so the insights view always shows all 12 archetypes, output is validated with Pydantic and repaired when malformed, and results are cached by input hash. GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; I designed the RAG-based AI assistant and worked on the scoring algorithm and data preprocessing for a tool, built as FastAPI microservices with a React and TypeScript frontend, that ranks cancer cell lines by target inclusion and exclusion genes and shows the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, SageMaker), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, Power BI, VS Code, Claude Code.
- **Languages:** Python (FastAPI, pandas, Pydantic, scikit-learn), TypeScript (React frontend, CellLineSelector), NodeJS (express, ReactJS), SQL, R (dplyr, tidyverse, ggplot2), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite